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Explainable Ensemble Machine Learning Framework with Optimized Feature Selection for Early Detection of Alzheimer's Disease

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***Abstract*— Alzheimer's affects millions of people, especially the elderly. Cases are supposed to be quadrupled by 2050. Early diagnosis is crucial for managing symptoms and delaying progression. Conventional diagnostic methods are expensive, time-consuming, and prone to errors. This study utilizes a Kaggle dataset of 2,149 records including 35 features to create a powerful machine learning model to identify the early signs of Alzheimer disease. Normalization, scaling, and label encoding were applied to improve model performance. These methods provided a consistent feature contribution and converted categorical data into numerical representations for faster learning. The class distributions were balanced using EasyEnsemble, with 20% going toward testing and 80% toward training. Recursive feature elimination and mutual information were used to choose the top 10 features. The results of every model were rated by its F1-score, recall, accuracy, and precision. Many models were employed, including Random Forest, AdaBoost, LightGBM, CatBoost, Gradient Boosting, and MLP. CatBoost achieved 96.74% accuracy, while hard voting ensembles improved to 97%. SHAP and LIME were employed to elucidate the model's predictions. This approach removes issues with interpretability, imbalance, and inconsistent results, in contrast to previous studies. Because of its accuracy, robustness, and explainability, the proposed machine learning approach raises the standard for AD prediction.**

Keywords*— *Alzheimer's Disease, Normalization, Scalling, EasyEnsemble, Mutual Information(MI), RFE, CatBoost, LightGBM, SHAP, LIME

I. Introduction

A long-term brain condition called Alzheimer's disease (AD) causes delusions, psychosis, and short-term memory loss that are sometimes confused with the symptoms of aging or stress [1]. AD must be managed with ongoing medication [2]. Since it is chronic, AD might persist for years or every moment of someone's life [3]. Approximately, there are 50 million victims of Alzheimer disease in the world and the trend forecast is that this number will double every five years, hitting approximately 152 million AD sufferers by 2050 [4]. In order to avoid major brain damage, it is essential to prescribe medication at an appropriate period [2]. Early illness detection requires a lot of data gathering, complex prediction algorithms, and the knowledge of a qualified medical professional, all of which add time and expense to the process [5]. More than three-quarters of dementia victims are low-income countries. The disease primarily affects elderly people [6]. Individuals may suffer from anxiety or memory issues, including forgetting names and places [7]. Close friends and family members may also observe that they have trouble remembering their names [8]. Patients with Alzheimer's disease may experience challenges like memory loss, time/place confusion, regular tasks becoming challenging, difficulties in problem handling, misplacing items, and poor judgment at this stage of the illness [9]. By conducting a comprehensive medical examination, a medical professional might identify a patient's concentration and memory difficulties [10]. Over the last couple of years, machine learning methods have had an increasingly significant impact on the healthcare industry [11]. There is a growing need to develop effective early diagnosis and treatment methods for Alzheimer's because there is presently no known cure [6]. We came to the conclusion after reviewing the relevant research papers that researchers have effectively utilized a number of algorithms for machine learning and that most of the studies focused on a number of metrics such as accuracy and precision [12]. Many machine learning systems exist but they are not reliable and inaccurate in predictions[13]. They also struggle with underfitting and overfitting [12]. This study uses a Kaggle dataset with 2,149 records and 35 characteristics to propose an accurate and comprehensible ML model for early Alzheimer's identification. The target variable was determined to be the diagnostic column. A strong preprocessing pipeline was put in place to maximize model performance. This pipeline included label encoding, scaling, and normalisation to lessen feature bias and increase learning effectiveness. To guarantee accurate assessment, the data was split into 80% training and 20% testing. The Easy Ensemble approach was used to address class imbalance, improving the

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model's capacity to learn from underrepresented classes. To increase efficiency, the top 10 features were found using a hybrid feature evaluation method that included mutual information (MI) and RFE. Precision, recall, F1-score, and accuracy were used to assess a number of models, including Random Forest, Gradient Boosting, CatBoost, LightGBM, AdaBoost, and MLP. A hard voting ensemble(CB-LGBM VoteNet) increased the accuracy to 97%. In order to provide model transparency for clinical decision-making, feature impact was explained using SHAP and LIME.

- Strong preprocessing (encoding, scaling, and normalisation) and EasyEnsemble (to solve class imbalance) are used on a 35-feature Kaggle dataset.
- Among the models examined, a hard voting ensemble had the highest accuracy of 97% using a hybrid feature selection (RFE + MI).
- To demonstrate feature influence and aid in clinical decision-making, SHAP and LIME were used to guarantee interpretability.

II. RELATED WORKS

Using nine ADNI dataset features, Velazquez et al.[14] predicted Alzheimer's disease. They trained a Random Forest (RF) model using an oversampling technique and attained an accuracy of 93.6%. The study had several restrictions, though. Important performance measures, including precision, recall, or F1-score, which are vital for evaluating model dependability, especially in imbalanced datasets, were not reported by it. Furthermore, the feature selection process was not clear, which raises questions regarding the generalizability of the model and possible exclusion of other informative features. Using autoencoders, Martinez-Murci et al. [15] proposed a deep learning method for Alzheimer's disease prediction. The model sought to learn significant patterns using manifold representations on MRI images, where features were extracted to reflect cognitive impairments. The work examined how every coordinate in the latent space of the autoencoder affected brain function interpretation. Furthermore, the study might not be clinically relevant if it lacked a comprehensive assessment of how well the acquired characteristics generalize over various patient populations. Leong et al [16] selected important attributes from the OASIS-I cross-sectional dataset using feature selection. The dataset contained missing values in the amount of 5.63%, which were eliminated. With an accuracy of 94.39%, Random Forest offered the highest results. Silva et al. [17] showed how to use multi-site neuroimaging data to look at various neurological conditions. After employing synthetic data to investigate brain structural correlations across disorders and clinical cohorts, they used the methodology to multi-database research like ADNI. Shahbaj et al. [18] employed rule induction, Decision Trees, Naïve Bayes, K-Nearest Neighbours, and a linear model to predict Alzheimer's on the ADNI dataset. The linear model was able to reach the highest accuracy of 88.24%.Tatiq et al [19] According to their research, Alzheimer disease is preventable by emphasising on modifiable vascular risks factors since both types of dementia coexist in the brain and share some risks which are modifiable. Neelaveni et al [20] used psychological characteristics to present an approach for AD prediction. Classification was executed using two ML classifiers, support vector machine and DT, where the SVM classifier performed the best, with an accuracy of 85%. Kavitha et al.[11] used voting, gradient-boosting classifiers, SVM, DT, and RF to perform prediction. Using the median approach, the SES missing data was filled. With an accuracy of 86.92%, Random Forest achieved the best results. An automated Alzheimer's prediction model was developed by Menagadevi et al. [21] using MRI datasets. Two datasets were employed in the experiments: one from the Kaggle repository and the other from ADNI. Curvelet thresholding was used for image preprocessing, while octagon histogram equalization was used for image improvement. Extreme learning machines, KNNs, and SVMs were used to perform the classification. Using values of k 3, 5, and 10, the k-fold validation procedure was used to validate the results. Williams et al. [22] generated forecasts of cognitive performance using four alternative models—DT, SVM, convolutional neural network , and NB—based on demographic and neuropsychological data. Because the gaps created by the missing numbers were filled in with average values, NB's accuracy was maximum in this case. Antor et al. [13] employed logistic regression, DT, RF, and SVM algorithms for Alzheimer's early detection. For prediction, eight OASIS dataset attributes were used. All features, with the exception of socioeconomic status, had missing values removed. In this feature, the median was used to fill in the missing values. Data for testing and training were split 80:20. A problem with overfitting plagued Random Forest. With 92% accuracy, SVM offered the best results.

III. MATERIALS AND METHODS

The entire process, which may involve processing, analyzing, and result interpretation, from collecting data to final prediction, is presented in “Fig. 1”.

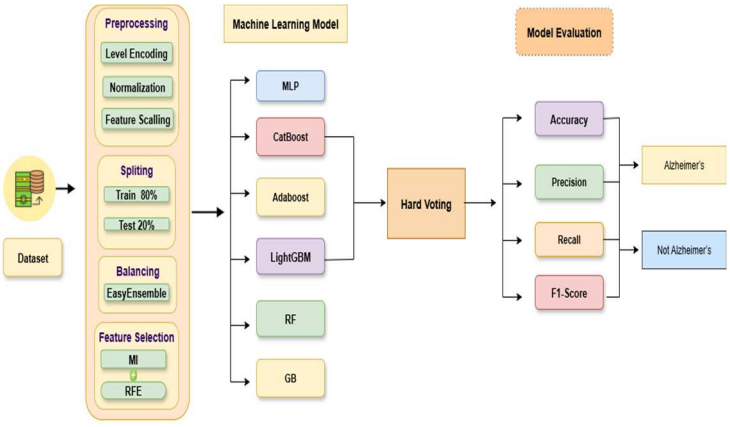


Fig.1. The proposed method for CB-LGBM VoteNet model

A. Dataset Description

This study makes use of the Kaggle [23] freely accessible Alzheimer's Disease Dataset, which comprises 2,149 instances and 35 parameters that cover a broad range of patient data. The features include demographic information (age, gender, and

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ethnicity), lifestyle factors (diet, physical activity, and alcohol consumption), clinical indicators (blood pressure, cholesterol, and MMSE scores), and cognitive symptoms (disorientation, forgetfulness, and confusion). The target variable is a column labeled Diagnosis.

B. Data Preprocessing

It was determined that there were no missing values in the dataset. Because of its low variability and predictive usefulness, the label-encoded category characteristic DoctorInCharge was eliminated. To guarantee an objective assessment, the data was divided 80:20 between training and testing. The target variable showed a significant class imbalance, which may skew model learning. In order to solve this, EasyEnsemble balance was only used on the training set, which increased the representation of minority classes while maintaining the original test distribution. This method guarantees an impartial, reliable performance evaluation while enhancing model generalisation.

C. Correlation Analysis

Significant correlations between important elements are revealed by the correlation matrix as shown in "Fig". 2. High blood pressure and hypertensive diseases are directly linked, as evidenced by the high positive association between systolic blood pressure and hypertension (r = 0.88). Memory complaints and MMSE scores have a negative connection (r = -0.76), underscoring cognitive loss as a key sign of AD. Furthermore, there is a moderately negative connection (r = -0.65) between physical activity and BMI, suggesting that more exercise lowers body mass index. The feature selection process was guided by these linkages, guaranteeing that the most pertinent properties were included for enhanced AD detection.

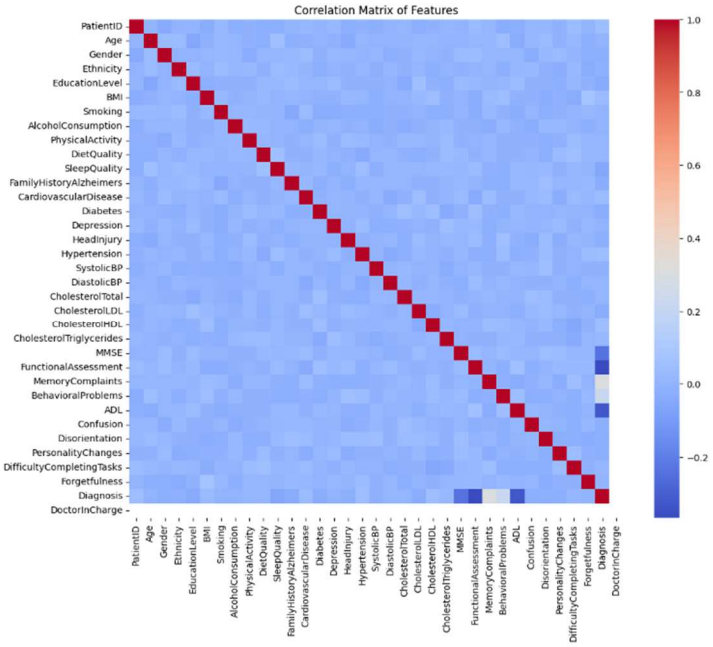


Fig.2.Correlation matrix of features

D. Feature Engineering

The goal of this study was to simplify the input space by choosing the top 10 most pertinent features, which were found using a two-stage hybrid feature selection procedure as illustrated in "Fig.3". Initially, each feature's nonlinear dependency on the target variable was measured using MI. The retention of the fifteen features with the best MI scores made sure that only the most instructive attributes were taken into account for additional refinement. To further reduce the dimensionality, an XGBoost classifier was used in conjunction with Recursive Feature Elimination (RFE). This method balanced model simplicity and prediction strength by recursively eliminating the least influential features until the top ten variables were found. By concentrating on the most discriminative characteristics, optimizing the training procedure, and reducing noise from extraneous data, this hybrid strategy enhanced both performance and interpretability.

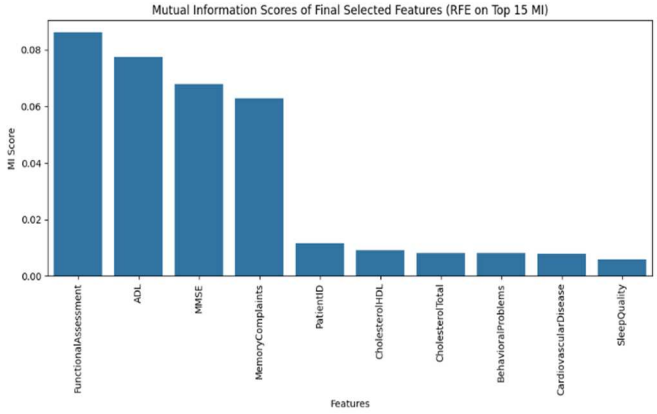


Fig.3. Top 10 selected feature visualization

E. Machine Learning Models

This study included six models: Multi-layer Perceptron (MLP) Classifier, Random Forest, LightGBM, CatBoost, Gradient Boosting, and AdaBoost. Because of its capacity to manage organized clinical data that combines categorical and numerical variables, each model was chosen. RF uses random selections of data to construct several decision trees, then uses majority voting to aggregate the results. This lessens overfitting and records intricate feature interactions. Gradient Boosting iteratively enhances performance by fixing the mistakes of the prior trees. LightGBM improves boosting with histogram-based feature binning and rapid, leaf-wise tree growth, which makes it effective for big datasets. In order to improve the categorization of more difficult cases, AdaBoost highlights previously incorrectly categorized data points and modifies the model's focus during iterations . An example of a neural network is the MLP Classifier, which uses layers of linked nodes to capture non-linear patterns. Despite its strength, its performance on structured datasets necessitates careful scaling and customization. CatBoost is a gradient boosting method that reduces overfitting and achieves high accuracy by handling categorical features efficiently with symmetric trees and ordered boosting.

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1) Voting: A voting classifier uses majority voting to predict a class by combining many models. Either soft voting or hard voting is used. To improve classification accuracy in this investigation, we employed hard voting with the top two models, CatBoost and LightGBM. More consistent and reliable predictions for Alzheimer's disease categorisation are produced by this ensemble strategy, which makes use of LightGBM's effectiveness on big datasets and CatBoost's prowess in handling categorical data.

In hard voting, the class with the most votes is chosen as the final output after the expected class labels from each of the base models are gathered(1).

$$\hat{y}(x) = mode\ (\ \hat{y}_{CB}(\mathrm{x}), \hat{y}_{LGBM}(\mathrm{x})\ ) \qquad (1)$$

This formula ensures a more consistent and dependable classification by determining the final prediction $\hat{y}(x)$ based on the most frequent output among CatBoost $\hat{y}_{CB}$ and LightGBM $\hat{y}_{LGBM}$.

F. Environment setup

The utilization of specific resources was necessary for this study. The proposed model is based on the following sources, which are shown in Table I.

TABLE I. ENVIRONMENT SETUP

Environment Setup	
Resource	**Details**
CPU	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.40 GHz
RAM	8.0 Gb, Speed 3200MHz
GPU	Intel®iris® Xe Graphics
Experimental Tool	Jupyter Notebook

IV. RESULTS AND DISCUSSION

A. Confusion Matrix

Confusion matrix is a fundamental assessment technique for assessing the performance of a classification model by comparing actual and predicted values. It shows where the model performs well and where it falls short by breaking down its predictions for the positive and negative classes. The matrix is divided into four primary sections: True Positive, True Negative, False Positive, and False Negative. This total accuracy is seen in (2). The ratio of genuine positives to anticipated positives, or precision, is computed using (3). The capacity to identify true positives is measured by recall, as shown in (4). F1-score is provided in (5), which is the harmonic mean of accuracy and recall.

$$\text{Accuracy: } \frac{(TP+TN)}{(TP+FP+FN+TN)} \qquad (2)$$

$$\text{Precision: } \frac{TP}{(TP+TN)} \qquad (3)$$

$$\text{Recall: } \frac{TP}{(TP+TN)} \qquad (4)$$

$$\text{F1-Score: } \frac{2 \times \text{Precision} \times Recall}{\text{Precision} \times \text{Recall}} \qquad (5)$$

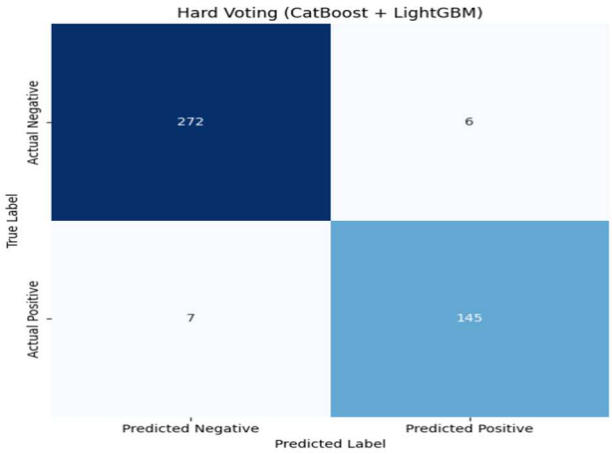


Fig.4. Confusion metrics for CB-LGBM VoteNet model

Using the confusion matrix, each class can be compared against the other class to identify the number of samples that were incorrectly classified. The best models' performance with the confusion matrix is displayed in "Fig.4".

B. Result Analysis

Table II indicates that LightGBM and CatBoost had the best single-model performance, at about 96%. Random Forest and Gradient Boosting were closely behind. MLP had the worst result, whereas AdaBoost had a mediocre one. With the highest accuracy of 97%, the proposed CB-LGBM VoteNet demonstrated the efficacy of the ensemble method.

TABLE II. EVALUATION OF DIFFERENT MODELS

DIFFERENT ML MODELS				
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
CatBoost	96.7%	0.948052	0.960526	0.954248
LightGBM	96.5%	0.936306	0.967105	0.951456
Gradient Boosting	96%	0.935484	0.953947	0.944625
Random Forest	95.8%	0.946667	0.934211	0.940397
AdaBoost	91.6%	0.871795	0.894737	0.883117
MLP Classifier	81.8%	0.780303	0.677632	0.725352
CB-LGBM VoteNet model	**97%**	**0.9602**	**0.9539**	**0.9570**

Table III shows that Antor et al. [13] obtained 92% accuracy with SVM on the exact same dataset, but Kavitha et al. [11] obtained 86.92% accuracy with RF on the OASIS dataset.Leong and Abdullah et al. [16] achieved 94.39% accuracy with RF on the OASIS-I dataset, whereas Velazquez

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et al. [14] achieved 93.6% performance with RF on the ADNI dataset.The suggested CB-LGBM VoteNet has the maximum accuracy of 97% and strong precision and recall (0.9602 and 0.9539), outperforming previous Alzheimer's disease detection approaches.

TABLE III. COMPARING PERFORMANCE WITH PREVIOUS WORK

COMPARING PERFORMANCE TO PREVIOUS WORK					
Authors	**Dataset**	**Methodology**	**Accuracy**	**Precision**	**Recall**
Kavitha et al. [11]	OASIS	RF	86.92%	-	-
Antor et al. [13]	OASIS	SVM	92%	-	-
Velazquez et al[14]	ADNI	RF	93.6%	-	-
Leong and Abdullah et al[16]	OASIS-I cross-sectional	RF	94.39%	-	-
Proposed CB-LGBM VoteNet	**Alzheimer's Disease Dataset(Kaggle)**	**CatBoost + LightGBM**	**97%**	**0.9602**	**0.9539**

C. AUC-ROC Curve

To determine the ability of the model to distinguish between positive and negative classes, the ROC curve that is the plot of the True Positive Rate (TPR) versus the False Positive Rate (FPR) at the various thresholds were plotted. We used 1111 negative and 608 positive data points to train the LightGBM model. Strong classification ability, which reflects the model's dependability in identifying Alzheimer's patients, is shown by a high AUC value. This demonstrates how well and resilient the model is at handling binary classification problems as portrayed in “Fig. 5”.

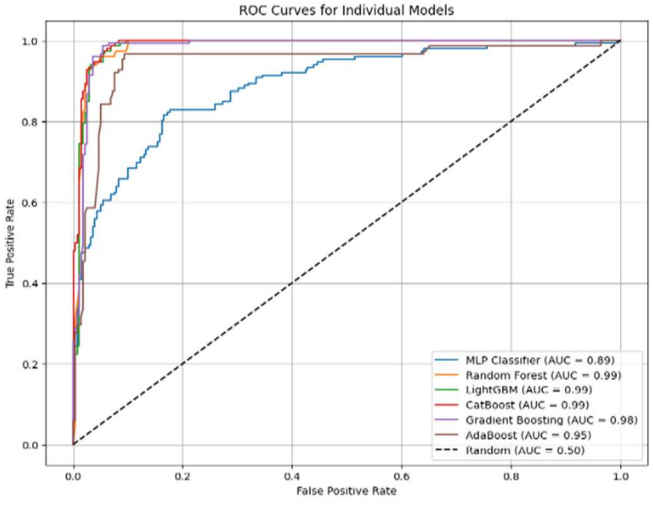


Fig.5. ROC Curve for Individual Models

D. Explainability Analysis

This section covered the model's performance as well as the factors that influence the model's prediction of a particular class and the values that produce this outcome.The model uses SHAP, LIME, and partial dependency to explain its prediction.

1) SHAP: The trained model's feature contributions were interpreted using SHAP. ADL, MMSE, and functional assessment had the most effects, followed by behaviour issues and memory complaints. Conversely, diastolic blood pressure, diet quality, and sleep quality had no effect. This emphasises the importance of the cognitive and functional tests in identifying Alzheimer’s, as presented in “Fig.6”.

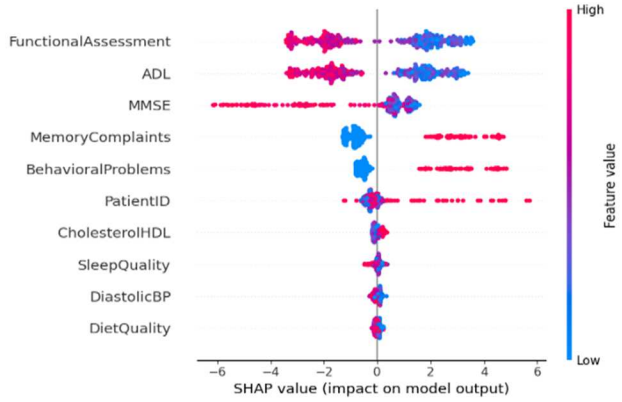


Fig.6. SHAP Value for CB-LGBM VoteNet Model

2) LIME: LIME uses smaller, interpretable models to locally approximate larger models, explaining how a model produces certain predictions. As seen in “Fig. 7”, the ensemble model in this study had a 100% confidence level in anticipating AD. According to LIME, characteristics like low MMSE, ADL, and Functional Assessment scores were significant predictors of AD. On the other hand, the lack of behavioral issues and memory complaints had a marginally negative impact. Dietary quality, diastolic blood pressure, and sleep quality also had a small impact. LIME improves interpretability and fosters confidence in AI-based medical diagnosis by coordinating model decisions with significant clinical indicators.

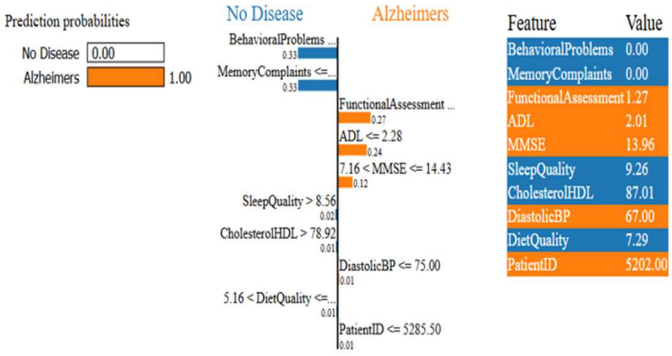


Fig.7.LIME Explanation of CB-LGBM VoteNet Model

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V. Conclusion and Future Work

In this study, a dependable and comprehensible ML model for Alzheimer's disease diagnostics is introduced. The suggested system obtained outstanding prediction performance by utilizing a well-preprocessed Kaggle dataset, using efficient feature selection techniques, and using EasyEnsemble to handle class imbalance.CatBoost outperformed findings from previous studies by achieving the highest single accuracy of 96.74% among the models studied. This accuracy was further enhanced to 97% utilizing a hard voting ensemble with LightGBM. Additionally, the model's transparency was enhanced by the combination of SHAP and LIME, which allowed for meaningful prediction interpretation a crucial component of medical decision-making. In contrast to earlier studies, this methodology successfully tackles important drawbacks such as inconsistent classification accuracy, unbalanced data, and unexplainable models. In terms of precision, resilience, and interpretability, the suggested model establishes a new standard for machine learning-based Alzheimer's prediction.

In order to improve generalizability, the model can be improved for subsequent research by adding more varied and longitudinal information. Since the current findings are based on a synthetic Kaggle dataset of 2,149 people from a single public source who are between the ages of 60 and 90, external validation on bigger, real-world multi-center cohorts is still crucial. Furthermore, broadening the application of explainability strategies like Anchors, ALE, and Counterfactual Explanations may enhance user confidence and clinical applicability. Creating a real-time diagnostic system (such as a web or mobile application) would enable prompt intervention and early risk identification, which could enhance results in actual healthcare settings. Additionally, the suggested framework can be included into already-existing diagnostic platforms, such as electronic health record systems, where it might operate on regularly gathered data to produce early risk scores and brief explanations of the primary contributing elements.

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